

On Datacenter Performance and Climate-Impact with Multi-Model Simulation and Analysis

Abstract—Datacenters are vital for the digital society but represent a considerable fraction of global energy consumption. To improve their sustainability and performance when demand is foreseen to increase, we envision simulators and simulation-based digital twins will become primary decision-making tools. However, unlike other fields focusing on key societal infrastructure such as waterworks and mass transit, datacenter simulators cannot yet combine multiple, independent models into their operation. Addressing this challenge, in this work we propose M3SA, a datacenter simulation and analysis framework that uses discrete-event simulation to predict, per model and then combined into a meta-model, the impact on climate and performance of various realistic datacenter conditions. We design an architecture for simulating multiple concurrent models, a technique to integrate the results of multiple models into a meta-model, and a procedure to evaluate the accuracy of the meta-model. Through experiments with a prototype, we show that (i) M3SA can be used to reproduce peer-reviewed experiments, and enhance their output with more diverse metrics and more detailed analysis; (ii) M3SA can be configured with a variety of realistic parameters, such as diverse workload traces (using Grid Workloads Archive data in our experiments), and energy production data such as carbon intensity over time and location (gCO₂/kWh data across all EU regions, from ENTSO-E); (iii) M3SA enables various types of what-if and how-to analysis, such as how to configure CO₂-aware migration over yearly energy-production patterns. M3SA has been integrated into the open-source software simulator OpenDC and is available on GitHub¹.

Index Terms—datacenters, multi-model simulation, performance, energy utilization, climate impact, sustainability, OpenDC

I. INTRODUCTION

Our society and economy are increasingly dependent on digital services. Correspondingly, new datacenters are being built, and existing datacenters are scaled up [1]–[4]. Before building or expanding, operators use simulators to project infrastructure performance and climate impact [5]–[7]. For example, in the EU research project Graph Massiziver, simulators predict speedup, failure cost, energy consumption, and CO₂ emissions for massive-scale infrastructure [8]. New models are developed or trained for specific scenarios, at significant cost and time [9]–[11]. However, datacenter simulation remains challenging due to the unpredictable behavior of datacenter software and hardware ecosystems [12], [13] and the continuous emergence of new devices and applications [3], [4]. Simulations based on *single* models are prone to errors when encountering edge cases and, as this work also showcases, new scenarios. Yet, current simulators, such as the open-source

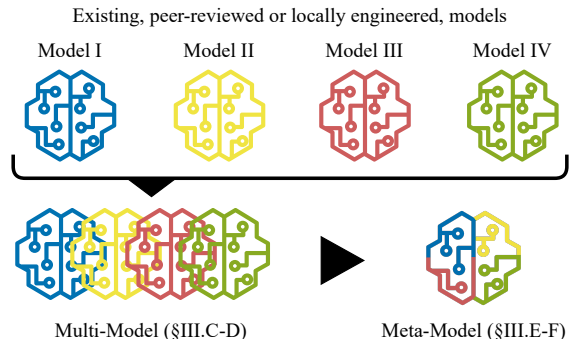


Fig. 1. M3SA multi-model simulation uses multiple models simultaneously. M3SA meta-model integrates all results. Figure 9 depicts a specific scenario.

OpenDC [14], CloudSim [15], [16], and SimGrid [17], [18], support only single-model simulation. Addressing this gap, in this work we design, prototype, and evaluate the M3SA (read as [mesa]) framework for *multi-model* (and meta-model) simulation and analysis of datacenters.

Figure 1 illustrates our approach. M3SA takes in existing models, e.g., peer-reviewed or engineered by the datacenter operator. Then, M3SA conducts multi-model simulation, automating the independent execution of each model. Last, M3SA integrates the results into a meta-model.

Open challenge: The lack of simulation capabilities to consider multiple models simultaneously and integrate their results into a meta-model. Albeit essential, singular models are insufficient for reliable predictions. A singular model makes accurate predictions in the context it has been trained for and often fails to handle real-life edge-case scenarios [14], [19], [20]. Moreover, singular models are calibrated to ideal scenarios and may lack adaptability to different *operational phenomena*, such as sudden spikes in demand, performance variability, equipment failures, and power outages. Such scenarios can lead to significant discrepancies between model-prediction and reality, leading to costly downtimes [21], impactful outages [22], and datacenter shutdowns [23].

We posit that combining multiple models can help alleviate the biases of each model, by contrasting their predictions for the same simulated scenario. The community leverages diverse modeling approaches trained or validated on diverse datasets [10], [11], [14], [24]–[26]—each an opportunity to include in multi-model simulation. However, none of the open-source datacenter simulators, including the highly cited CloudSim [15], [16], SimGrid [17], [18], OpenDC [14], and GDCCSim [27], currently supports multi-model simulation.

¹URL for double-blind review: <https://anonymous.4open.science/r/m3sa/>

Simulating using multiple models is a novelty in computer systems; conducting multi-model simulations for datacenters is also innovative beyond computer systems science. Coarse-grained, large-scale models in weather and climate simulation [20], [28] and detailed, small-scale models in ecology [29] already use simulation instruments that rely on multiple prediction models, calibrated and adjusted for the needs of their scientific field. In contrast, to obtain meaningful predictions about complex datacenters ecosystems, medium-scale models must combine higher-level abstraction with detailed operational models for specific devices and applications, and address various operational phenomena. This is fundamentally different than other-scale sciences [30].

Approach and main contribution: In this work, we address the open challenge by designing, implementing, and evaluating M3SA in complex, realistic, trace-driven simulations, with a three-fold contribution:

- C1 (Design)** We design M3SA, the first Multi- and Meta-model simulator for datacenters (Section III). We formulate functional and non-functional requirements. We propose an overall architecture for Multi- and Meta-simulation of datacenters. We further design the Multi- and Meta-Model components to work with generic discrete-event simulators, and propose assisted configuration and evaluation (validation) processes.
- C2 (Experimentation)** We prototype M3SA and conduct three large-scale, trace-driven datacenter experiments (Section IV). We prototype M3SA and embed it into OpenDC [14], a peer-reviewed, state-of-the-art datacenter simulator. The three experiments: (i) Reproduce a peer-reviewed, open-science datacenter experiment [9] (Section IV-B); (ii) analyze the impact of workload kind—scientific [31] vs. business-critical [32]—on datacenter performance and CO₂-emissions (Section IV-C); and (iii) analyze the impact of EU-region on the performance and CO₂-emissions of migrated workload (Section IV-D). Overall, M3SA enables the analysis of both datacenter performance and CO₂-emissions related to resource use, across diverse workloads and geographical locations, with improved accuracy and explainability over single-model simulations.
- C3 (Open science)** We contribute to open science all the software and data [33] created in this project. We engineer the M3SA software and integrate it into OpenDC, which is based on the popular Kotlin and Python languages [34]. For the last experiment, we collect public CO₂-emissions data across all EU countries from ENTSO-E, align the timestamps, and compute carbon-intensity data from it, and release the resulting FAIR dataset. Last, we make available the complete reproducibility capsule: <https://anonymous.4open.science/r/m3sa/>.

II. BACKGROUND

We present a background on datacenter simulators, prediction models, and metrics used in this research and widely used in the community.

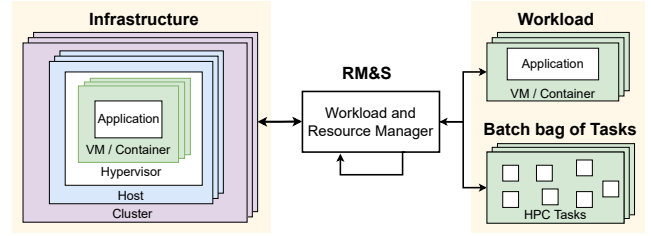


Fig. 2. Simulation enables fine-grained infrastructure operational monitoring used by stakeholders in decision-making processes. Adapted from [37].

“Simulation: Simulation is defined as the imitation of the operation of a system or real-world process over time, and in many cases, manufacturing provides one of the most important applications of simulation” [35]. Simulation provides datacenter stakeholders with operational insights into the behavior of digitally twinned infrastructure running under a given workload. *Workloads* contain tasks operated on physical machines, virtual machines (VM), or containers. *Traces* represent recordings of real-world events, capturing detailed operational data of the infrastructure under given workload(s); traces provide a granular view of resource usage, essential for driving simulations or replaying real-world scenarios.

Predictive models in large-scale computer systems are empirical prediction systems that analyze, combine, and compute various input elements (often, traces) to produce fine-grained output predictions [36, §1]. We use such models to predict how real-world workloads run on ICT infrastructure, under various operational phenomena, as modeled by users. In this work, we use three types of models: (i) *Energy models* predict the amount of energy the datacenter consumes from the power grid. (ii) *CO₂ emission models* predict the carbon intensity (CO₂ emitted per unit of energy, measured in gCO₂) of the datacenter under the simulated workload. (iii) *Failure models* simulate task failures according to a failure trace. Without such traces, failure models generate task failures with arbitrary but realistic occurrence timestamps and recovery periods.

We use *metrics* for system monitoring and accuracy measurement. For system monitoring, we use two main metrics: *energy consumption* and *carbon intensity*, measured in international system units. We define and measure the models’ accuracy against real-world data (i.e., measured reality) using *Mean Percentage Absolute Error (MAPE)*, expanded in Section III-F.

III. DESIGN OF M3SA: A TOOL FOR MULTI- AND META-MODEL SIMULATION AND ANALYSIS

In this section, we synthesize requirements and design a novel approach for datacenter simulation. We propose M3SA, a system that enables simulation using *multiple* models.

A. Requirement Analysis

(FR1) Leverage multiple models into a unified tool: The system must enable combining multiple (existing) models into a unified tool. It should enable visualizing their predictions, independently or combined, thus increasing the explainability of predictions, and pointing out idiosyncratic errors or biases.

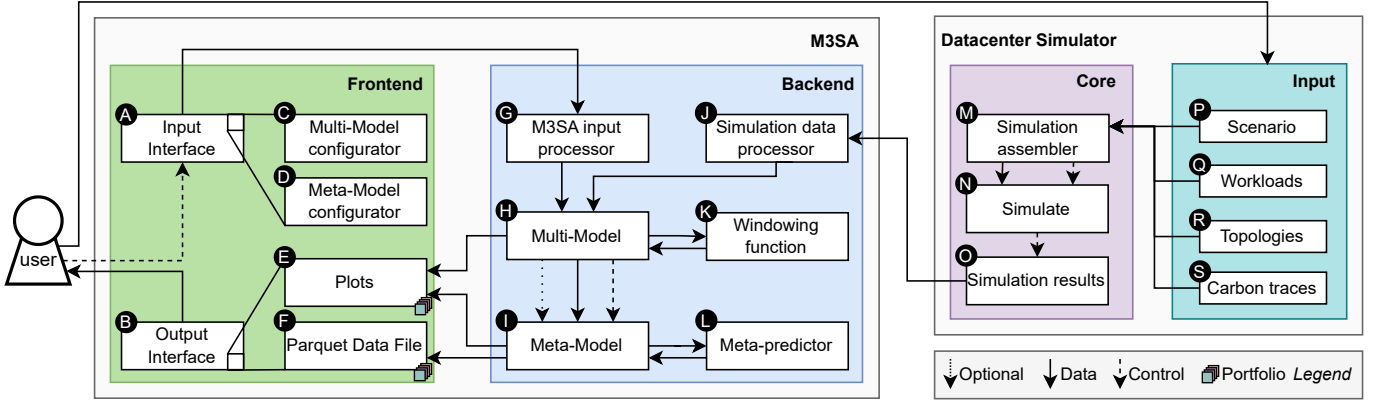


Fig. 3. Architecture overview of M3SA, a Multi- and Meta- Model Simulation Analyzer, integrated with a black-boxed simulator.

(FR2) Predict by integrating predictions of multiple models: A *Meta-Model*, able to aggregate single-models' predictions, at a user-selected granularity, to alleviate errors and biases of idiosyncratic models, thus strengthening the credibility of the predictions.

(FR3) Open-source portable tool: The system should be open-sourced, and usable either decoupled or coupled with a simulator. Porting to another simulator should be convenient.

(NFR1) Provide in-meeting or same-day simulation results: We assume the multiple models can be run in parallel, reducing the simulation time to a matter of minutes or less². The analysis part of the system should only add a fraction to the time needed to run the simulation itself, and not more than double it.

(NFR2) Highly accurate predictions: The *Meta-Model* should predict with higher accuracy than the average individual model employed in the simulation. Equivalently, the error rate of the *Meta-Model* should be lower than the error rate of the average model.

(NFR3) Operation at real-world scale: All FRs and NFRs should be met for *at least* 8 individual models, on input traces of *at least* 200,000 samples per model or 2 years at the industry sample rate of 5 minutes [14], [37].

B. Overview of the M3SA Architecture

We design M3SA to be capable of operating coupled or decoupled from a datacenter simulator. Figure 3 depicts an overview of the system's architecture, in which M3SA extends a black-boxed simulator (**FR3**). Throughout this paper, we couple M3SA with OpenDC, a peer-reviewed, open-source, discrete-event simulator with simple interfaces, and over 5 years of development and operation [1], [14].

M3SA adopts the architecture we define as “Simulate First, Compute Later” (SFCL), where M3SA operates only after the simulator successfully predicts and the *simulation results* (O) are available. This architecture couples M3SA to the simulator as a top, external layer. We also considered the alternative architecture of “Compute While Simulating” (CWS), where

M3SA would be deeply embedded into the simulation infrastructure. Comparing SFCL and CWS architectures, we observe a tradeoff between the depth of M3SA embedding (**FR3**) and the performance of the overall system (**NFR1**) - the more embedded M3SA is, the better the overall performance, yet the more complex it becomes to port M3SA from one simulator to another. We choose the SFCL architecture, thus allowing simple adoption of M3SA; to meet the performance requirements, we focus on reducing redundancies and following good engineering practices.

In the SFCL architecture, the user interacts with the system through the *Input Interface* (component A) and *Output Interface* (B) interfaces (**FR3**). The M3SA process begins from the input, where the users configure the *Multi-Model* (C) and the *Meta-Model* (D). The simulation process, triggered and controlled by the M3SA backend, happens between M-S: the system sets up a simulation based on user-input, simulates, and centralizes predictions. The simulation block M-S corresponds to how discrete-event simulators commonly used in the field actually operate, and closely matches the OpenDC [14] and CloudSim [16] architectures; in particular, the simulation assembler (M) is where single models are typically defined in current experiments.

Further, the M3SA backend retrieves the predictions made by the simulation infrastructure, one per single model, and assembles the *Multi-Model* (**FR1**). To avoid redundant computation, the *Meta-Model* predicts using already leveraged and processed data from the *Multi-Model* (**FR2**), (**NFR1**). Lastly, the system plots the predictions of the *Multi- and Meta-Model* (E) and outputs the *Meta-Model* predictions to a lossless-compressed data file (F).

C. Design of the Multi-Model Component

In this section, we define explainability, and elaborate on the *Multi-Model* and *Plots* components (Figure 3 H and E, respectively). The M3SA multi-model approach helps compare the outputs individual models, thus improving users' understanding of the predictions and increasing explainability (**FR1**). It also enables meta-modeling, as detailed in Section III-E.

²Complex datacenter experiments using OpenDC can be used as a reference point. They are reported to take seconds to minutes [14].

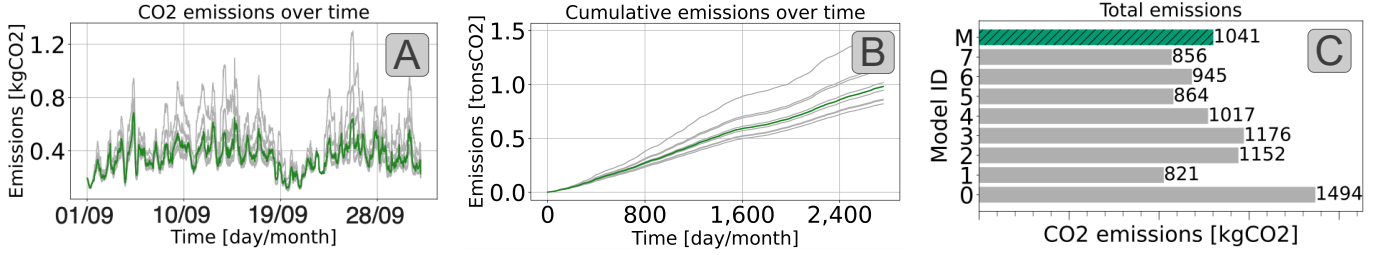


Fig. 4. Sample M3SA plots. We represent individual models in gray and Meta-Models in green. In sub-plots A and B: vertical axis depicts the simulated metric, horizontal axis depicts time. In sub-plot C: vertical axis depicts model identifier, horizontal depicts cumulative totals.

We define *explainability* as the user’s understanding of the possible behavior, limitations, and biases of the system under test, as predicted by relevant, peer-reviewed or locally vetted, models. Better explainability is achieved when multiple models are available, and when at least the ranges of acceptable predictions, and also the detailed individual predictions, are available and contrasted across the models, enabling specialists to better reason about the problem³. Ideal explainability both matches measurable reality and links real-world values to predictions made by one or several models, thereby enabling specialists to explain wherever evidence exists about which model gives better predictions and how close those predictions are to instrumental observations (traces)—proxy to reality.

The Multi-Model component (H) leverages multiple models into a unified tool. Within this component, predictions of singular models, which M3SA has already produced through its Datacenter Simulator backend, are centralized; the user can select the form of data representation. To increase performance and alleviate redundancy, the system reads from the input data file only the metrics selected by the user (typically, at configuration time) and discards irrelevant data. The Multi-Model process then filters prediction data from individual models with a user-selected windowing function, which minimizes the dataset for better performance (NFR1) and also removes noise for better visual comprehension; we expand on the concept of windowing in Section III-D. After completion, the Multi-Model is ready for further operations, such as Meta-Model computation or plotting (FR2).

The Plots component (E) uses the Multi-Model results to produce, guided by typical user-given configuration, visual outcomes as in Figure 4. M3SA supports analyzing how metrics evolve over time, enabling the reader to analyze predictions at specific timestamps (Figure 4A), how quantities accumulate over time (Figure 4B), and aggregates such as sums (totals) over a part or the entire experiment (Figure 4C).

D. Analysis and Configuration of Multi-Model Window Size

To address the tradeoff between the detail of the visual comprehension (FR1) and the performance (NFR1) aspects of the multi-model, we equip the Multi-Model component with a windowing mechanism.

³In many scientific domains, e.g., in weather and climate modeling [20], [28], multiple instrumental observations (traces) and multiple models are used to provide evidence, toward building sufficient confidence in specific outcomes.

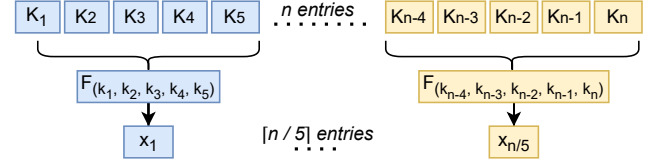


Fig. 5. Window size 5 applied on n entries, using aggregation function F .

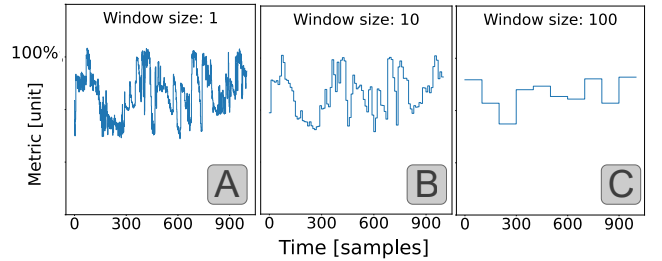


Fig. 6. Comparison of visual quality when windowing representative input data. Window sizes differ per sub-plot, in order, 1, 20, and 100. The horizontal axis represents time, in abstract units. The vertical axis represents the system response metric, from zero to maximum.

We define a *window* as an aggregation of chunks of data into a singular data entry, similar to a one-dimensional convolutional layer. Figure 5 illustrates how a window of size m (here, $m = 5$) is applied to n data entries. Each group of m elements is fed into an aggregation function F , leading to a final compressed output array of size $\lceil n/m \rceil$.

We now explore the trade-off between visual quality and performance. We conduct experiments with actual outputs produced by the simulator, for input between about 2,000 and 200,000 samples long; with various window sizes between 1, corresponding to retaining all data, and 1000, a large value; and without M3SA-level optimizations such as eliminating redundancies. (When optimizations are enabled, as indicated by the results in Section IV-B, the performance of M3SA becomes compatible with (NFR1).) Figure 6 presents a representative selection of the results. As expected, the larger the window size, the sparser the filtered output and thus the coarser the visual detail. At a window size of 100, and beyond, the actual shape of the input is irretrievably lost. On the other side of the trade-off, the parsing and graphing sequence of our largest test-input ($n = 201,600$), when $m = 1$ requires, on a common, off-the-shelf, Apple M2 MacBook Pro, about 4.5 minutes (264.7 s). As the window size filters out data to persist and graph, the runtime of computation and graphing decreases as the window size increases, to just over 3 minutes at $m = 10$. From $m = 20$, additional effects, e.g., the working set becoming increasingly large, reverse this trend, and the

time needed when $m = 1,000$ for the largest input already approaches the time taken without windowing (238.3 s).

Based on these experimental results, we configure the Meta-Model with a windowing size $m = 10$ for this work.

E. Design of Meta-Model component

In this section, we detail the *Meta-Model* component (Figure 3, ①). Overall, the Meta-Model uses an automated process to combine predictions made with multiple models (**FR2**) into a highly accurate result (**NFR2**).

The Meta-Model relies on the predictions made by multiple models. For performance reasons (**NFR1**), this process leverages processing results from the Multi-Model (see Section III-C), thus avoiding redundant computation. Upon initialization, the core of the Meta-Model uses the Meta-predictor (①), which combines other models' (windowed) predictions (**FR2**). After ① completes, the Meta-Model contains a one-dimensional set of predictions, prepared for output.

The output involves two independent processes, for usability (**FR1**), performance (**NFR1**), and portability (**FR3**) reasons. First, the system compresses the predictions losslessly, into a Parquet data file, which the user can process outside of the M3SA system (②); we use Parquet due to the scalability, efficient storage, and cross-system compatibility (and thus portability) of this storage format [38]. Second, the Meta-Model is plotted (③) following user inputs (④).

We define *robustness* as the confidence that predictions perfectly and consistently match the measured reality. Although the *accuracy* of the Meta-Model is directly dependent on the accuracy of individual models (①) and on the aggregation function (②), the robustness of the Meta-Model accuracy should intuitively be higher than of any single model (**NFR2**), because the Meta-Model can mitigate the errors of individual models when it combines the results across multiple models. This can be expressed and empirically measured for specific inputs, e.g., the MAPE metric introduced in Section III-F.

To achieve high accuracy, the Meta-Model component uses the Meta-predictor (①). The Meta-predictor receives a set of predictions over time, one per input model, with time divided in equal chunks (steps). As Figure 7 illustrates, the Meta-predictor first aligns the steps across all models, for example, removing entries when too few models provide predictions, and aggregates the remainder. In the figure, the rows represent the predictions of each of the m models, and the columns represent the aligned time-steps. Model 1 provides additional predictions, for time steps C_{n+1} and C_{n+2} , which are discarded by the Meta-predictor because not enough other models also provide predictions for these steps.

Following alignment, the Meta-predictor applies an aggregation function. In the figure, the values stored in each column C_k are aggregated for each time-step, using a (user-configurable) function F_k . In Figure 7, the aggregation function is applied vertically for each column, which is convenient also for performance. However, the concept also allows for diverse spatial combinations and further, more complex but model-order-free, approaches. We leave for future work the

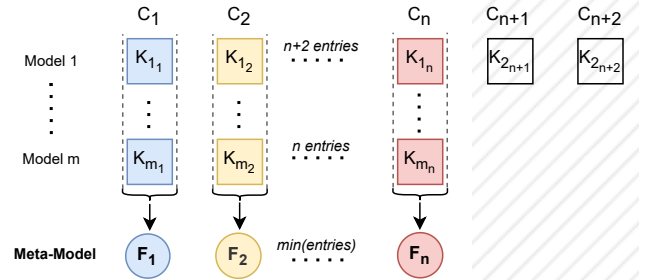


Fig. 7. The Meta-Model predictor, aggregating predictions of m models vertically, for each time-step C_1 to C_n . Steps C_{n+1} and C_{n+2} are discarded.

exploration of such aggregators, using mathematical, statistical, and machine-learning techniques.

For the aggregating function F , we explored for this work two computationally light, commonly used aggregation functions, (arithmetic) *mean*, which is used ecology [29] and meteorology [20] but has high sensitivity to outliers, and *median*, which alleviates the outlier sensitivity of the mean but is more computationally expensive.

The proposed Meta-Model design poses two challenges. Firstly, timestamp misalignments may occur due to non-deterministic simulation infrastructure. We address this challenge by selecting a datacenter simulator that establishes the same initial sampling timestamp for all models and the same sampling rate of the infrastructure. Secondly, individual models may output prediction datasets of distinct sizes (i.e., longer or shorter virtual infrastructure execution times), as we illustrate in Figure 7, for models 1 and m ; this difference may occur due to various reasons, such as scheduling algorithms that may prioritize tasks differently, or the simulation of potential infrastructure failures. To address this challenge, the meta-predictor uses only the minimum number of entries per model in the computation process.

F. Evaluation of Meta-Model Accuracy

To quantify robustness (see Section III-E) in M3SA (**NFR2**), we use the Mean Absolute Percentage Error (MAPE), a widely used relative-error metric [9], [39], [40]. MAPE equally penalizes positive and negative errors and is calculated using Equation (1), where n is the number of samples, R is real-world data, S is simulation data, i is the sample index:

$$MAPE [\%] = \frac{1}{n} \sum_{i=0}^n \left| \frac{R_i - S_i}{R_i} \right| \cdot 100 \quad (1)$$

Anticipating the need for further accuracy evaluation (**NFR2**), we design M3SA to be extensible to other metrics such as Normalized Absolute Difference (NAD) [9], Root of the Mean Square Error (RMSE) [41], and Mean Average Error (MAE) [42], [43].

IV. COMPLEX, TRACE-BASED EXPERIMENTS WITH M3SA

In this section, we conduct diverse experiments, evaluating M3SA and simultaneously providing new insights into datacenter operation. To setup our experiments (Section IV-A),

TABLE I

EXPERIMENT CONFIGURATIONS. (WT = WORKLOAD TRACE, CT = CO2 TRACE, PM = PERFORMANCE METRICS, SM = SUSTAINABILITY METRICS, IM = INDIVIDUAL MODELS, MW = MULTI-MODEL WINDOW SIZE, MF = META-MODEL AGGREGATION FUNCTION, SUO = SYSTEM UNDER OBSERVATION.)

Section	Focus	Input		Output					Failures	SUO
		WT	CT	PM	SM	IM	MW	MF		
§IV-B	Reproduce peer-reviewed experiment	WT1	✗	CPU hours	Wh	4	10	median	✗	S1
§IV-C	Analyze M3SA on fundamentally different traces	WT2, WT3	CT1	CPU hours	gCO2	8	1	mean	✓	S2
§IV-D	Adopt M3SA in CO2-aware, dynamic, workload migration	WT2	CT2	CPU hours	tonsCO2	16	1	median	✗	S3

TABLE II
SYSTEMS UNDER OBSERVATION (SUO).

ID	Source	Scale [#hosts]	Resources per host
S1	SURF	277	128 GB RAM, 16 cores, 2.1 Ghz
S2	Marconi	150	196 GB RAM, 48 cores, 2.1 Ghz
S3	Marconi	2,982	196 GB RAM, 48 cores, 2.1 Ghz

TABLE III

WORKLOAD TRACES USED IN EXPERIMENTS. NAME IS SOURCE AND COLLECTION YEAR (E.G., SURF-22 = SOURCE SURF, YEAR 2022). SUO = SYSTEM UNDER OBSERVATION, SR = SAMPLING RATE, CH = CPU HOURS (MILLIONS), S = SCIENTIFIC, B = BUSINESS-CRITICAL.

ID	Name	Type	SUO	Duration	Jobs	CH	SR
WT1	SURF-22	S	S1	7 days	7,850	0.31	30 s
WT2	Marconi-22	S	S2	30 days	8,316	4.74	20 s
WT3	Solvinty-13	B	S3	30 days	50	0.13	30 s

we first implement a prototype of M3SA, interface it with OpenDC, and provide the system open-source (**NFR2**). In Section IV-B, we reproduce a peer-reviewed experiment with M3SA capabilities and analyze M3SA against established requirements. In Section IV-C, we run an experiment using two fundamentally different traces, analyze the results, and evaluate the versatility of M3SA. Lastly, we evaluate M3SA on CO2-aware migration over energy-production patterns. We focus here on the main findings and further analyze the experiments in a technical report [44].

Overall, the experiments support four main findings (MFs):
(MF1) Overall, the Multi-Model can increase the explainability of the simulation results, and help in identifying biases and errors of single models; the Meta-Model can enhance prediction accuracy against single models, robustly. Specifically, the Meta-Model can reduce the error rate of the average individual simulation model by about 50% (MAPE decreases from 7.59% to 3.81%) with under 20% computational overhead (Section IV-B).
(MF2) M3SA can simulate 1 (or 2) years of datacenter operation in 5 (7) minutes, of which under 0.5 (1) minutes are M3SA overhead (Section IV-B).
(MF3) Supporting various C-level decisions, M3SA can simulate business-critical and scientific workloads in scenarios including machine failures (Section IV-C).
(MF4) Adding to MF3, M3SA can help in CO2-aware workload migration and scheduling processes, with decisions that

TABLE IV

CARBON TRACES USED IN EXPERIMENTS. NAME IS SOURCE, LOCATION OF COLLECTION, AND COLLECTION YEAR (E.G., ENTISOE-NL-22 = SOURCE ENTISOE, COLLECTED IN THE NETHERLANDS, YEAR 2022.)

ID	Name	Location	Duration	Sampling
CT1	ENTISOE-NL-22	The Netherlands	1 year	900 s
CT2	ENTISOE-EU-23	29 European regions	1 year	900 s

can help reduce CO2 emissions by 97.5% compared to the average location, and by 11% compared to the best but single location (Section IV-D).

A. Experiment Setup

We summarize our experimental design in Table I, and detail the systems, workloads, and carbon traces in Table II, Table III, and Table IV, respectively. We detail the most important aspects:

Workload traces (all public): SURF-22 is a scientific workload trace used in peer-reviewed experiments [9]. SURF-22 was traced in one of the largest HPC facilities at SURF, the Netherlands. It traces scientific jobs run in production, batch, with an average duration of 39.52 CPU-hours, and also detailed energy use over the same period.

Marconi-22 is one of the most detailed open-source workload traces currently available. It traces scientific jobs, and also multiple operational layers and the electrical network of Marconi HPC, a powerful supercomputing facility hosted by CINECA in Italy [31].

Solvinty-13 is a business-critical workload hosted in a Solvinty datacenter [45], former BitBrains, a mid-sized cloud service provider in the Netherlands. Solvinty-13 has been used in highly cited, peer-reviewed publications on business-critical workloads [32]. The Solvinty-13 traces long-running jobs, with an average of 2,722 CPU-hours per job.

Carbon traces (public): ENTISO-E Transparency Platform [46] consists of 40 European Transmission System Operators (TSOs), providing open source datasets about energy and sustainability metrics since 2015. In this work, we use traces from 29 countries from the ENTISO-E datasets.

Failure model: The failure model causes infrastructure failures at arbitrary times, with downtime periods following Lsdns04 traces [47], which follow an exponential distribution with known parameters [47]. We assume no checkpointing

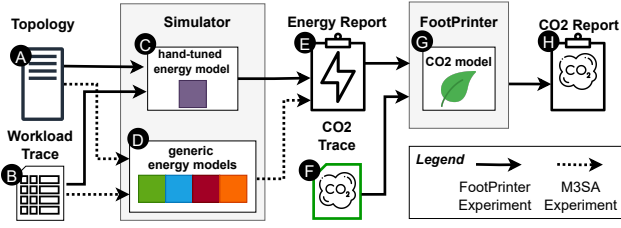


Fig. 8. Overview of experiment 1, adapted and re-run from FootPrinter [9], with M3SA simulation and analysis capabilities.

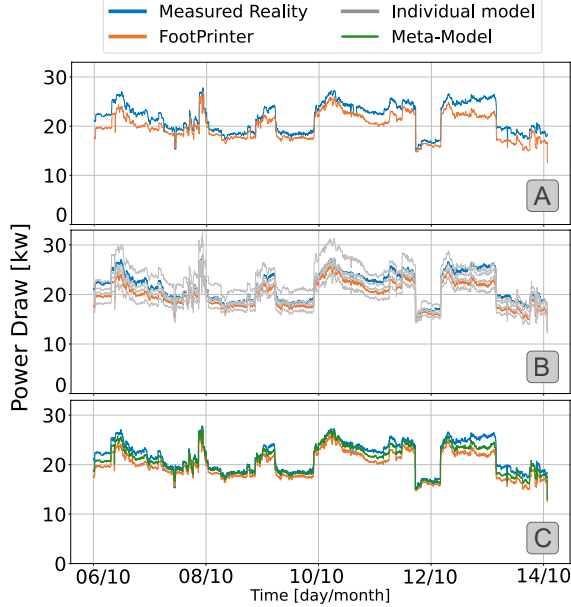


Fig. 9. Simulation results vs. measured reality: (A) FootPrinter simulator, peer-reviewed experiment [9]; (B) Multi-Model with four individual models vs. FootPrinter; (C) Meta-Model derived from the Multi-Model at point B.

system, so each job is re-run from the beginning when failures occur.

B. Peer-reviewed experiment reproduced with M3SA

Niwenhuis et al. propose FootPrinter [9], a tool for predicting the CO2 footprint of datacenters. As Figure 8 depicts, we reproduce here the experiment from [9], leveraging M3SA capabilities to achieve higher accuracy and more detailed explanations; Section IV-A and Table I summarize the design of this experiment. Illustrative for the differences between approaches, whereas Footprinter employs a single, hand-tuned energy model (C), M3SA uses multiple, generic energy models (D). The other components of the experiment, A, B, and E to H, remain unchanged, enabling a direct comparison between approaches. Our results strongly support MF1.

Figure 9 shows the power draw of the S1 system over one week. Figure 9A shows the results of reproducing the experiment from FootPrinter [9], compared against measured reality. Illustrative for a hand-tuned model, the FootPrinter curve approaches in many places the measured reality. As a trade-off, changing the model for a new setting would be complex and error-prone, requiring possibly a similar amount of work to the development of the initial model. (We explore this ability for M3SA, in Section IV-C.)

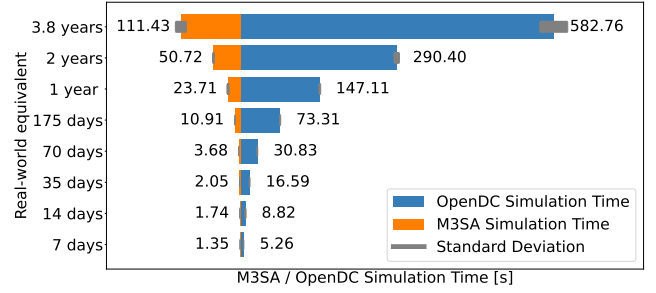


Fig. 10. M3SA performance overhead compared to OpenDC simulation runtime. The plot, which takes its structure from Tukey’s population pyramid, plots on the horizontal axis the system response: (left) M3SA’s overhead, (right) the OpenDC simulation time. On the vertical axis, discretely, is the duration of the workload; higher values cause higher runtimes.

Better explainability: Figure 9B additionally shows four individual models, whose results were automatically produced through the Multi-Model component. It seems appealing to select the most accurate model, from this sub-plot or peer-reviewed publications, by comparing against the ground truth provided by the measured reality. However, in real-world situations, and increasingly when opaque clouds provide resources, the analyst is often unaware of the ground-truth. The results in Figure 9B show that three of the individual models match the measured reality, while a fourth model constantly overestimates. Using the Multi-Model approach, the analyst can identify individual model biases by contrasting the predictions at each moment in time, thus increasing the overall prediction explainability⁴ over single-model approaches, where inaccuracies may not even be observable.

“All models are wrong, but some are useful” (George Box, 1976). Simulating with a robustly accurate model is a critical yet non-trivial challenge in ICT. Figure 9C shows the predictions of a Meta-Model, automatically derived from the Multi-Model presented in Figure 9B, and compared against the predictions of the FootPrinter model and the measured reality. We quantify accuracy against measured reality and obtain a MAPE of 7.59% for the average individual model and a 50% lower (better) MAPE of 3.81% for the Meta-Model. The improvement in error rate is due to the *median* aggregation function, which effectively filters extremes, reduces biases, and improves the robustness of the prediction.

Compared to FootPrinter’s MAPE of 3.15%, the Meta-Model has a higher error rate. However, FootPrinter underlies a manual simulation model, explicitly trained for a single trace (here, SURF-22). In contrast, M3SA enables the versatile, automatic re-use of generic models.

M3SA runtime overhead under realistic conditions: Throughout this experiment, M3SA operated on 20,170 samples, or about 7 days of operation with SURF monitoring rate of 30 seconds [14]. To assess the runtime overhead of M3SA, we further create datasets of various sizes, equivalent to different datacenter operational times, including 2 years (NFR3) and up to nearly 4 years, by modifying the monitoring granularity

⁴Resolving the discrepancies between models, and identifying a single best model, requires capabilities beyond the scope of this work.

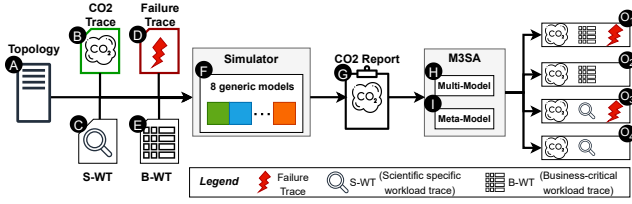


Fig. 11. Experiment 2 overview. Evaluating M3SA on business-critical and scientific-specific workloads, with operational phenomena.

of the simulator.

Figure 10 summarizes the results of this performance-analysis experiment. The M3SA overhead includes synthesizing a Multi-Model, computing the Meta-Model, and outputting the results. We run each experiment 10 times on a standard MacBook M2 Pro and represent the standard deviation in gray; full analysis is available in Appendix B [44]. On average, M3SA adds less than 20% time overhead to the overall simulation process, even for massive-scale datasets, and can simulate years of datacenter operation within minutes (NFR1).

C. Evaluating M3SA on fundamentally different traces

Support for NFR2 also introduces the question of which models can be considered. The ability to cope with the diversity of possible workloads is already understood as essential for datacenter simulators [48]. Similarly, failures are known to change how specific workloads are run [48], [49]. We showcase in this section, through an experiment, how M3SA copes with fundamentally different traces and with machine-failures. Section IV-A and Table I summarize the design of this experiment. As Figure 11 depicts, the experiment design includes two kinds of workloads to be run independently (C and E), and the option to include failures (D); these lead to four output combinations, O₁ to O₄.

Overall, besides the ability of M3SA to provide answers for these complex conditions (MF3), the results indicate that the presence of failures can add 21.9% CO2 emissions compared to running the workload without system failures, so sustainability studies such as [3] should explicitly include such analysis. Consequently, C-level stakeholders should not make decisions absent such analysis.

We run two fundamentally different traces. Not only do they come from different domains, scientific vs. business-critical, but also, whereas Marconi includes many job arrivals that fit diurnal and day-of-week patterns [31], Solvinity-13 has much longer jobs (see Section IV-A) and forms a stable workload that is largely time-insensitive [32].

Figure 12 shows the environmental impact of the two different workloads, when failures are not or also considered. The results support that M3SA can help identify (overly) biased models, even when considering only the aggregate results. In this experiment, we observe that model 0 constantly overestimates, predicting up to 44% more than the average of the other models. This major overestimation, albeit clear in a multi-model simulation, would not be observable in a single-model simulation; here, the Meta-Model approach helps.

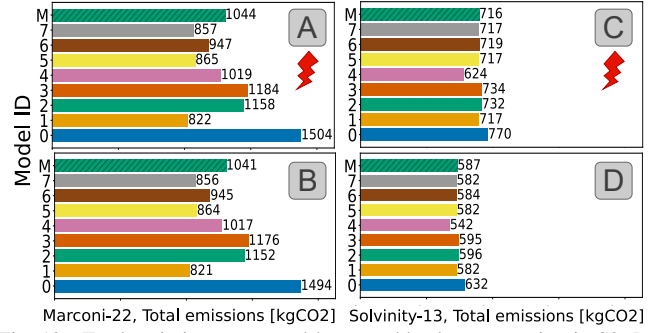


Fig. 12. Total emissions generated by a workload trace running in S2. Left column, Marconi-22 workload: (A) S2 experiences Lsdns04-based failures, (B) S2 without failures. Right column, Solvinity-13 workload: (C) S2 experiences Lsdns04-based failures, (D) S2 without failures. Each plot shows the total CO2 emissions predicted by the M3SA Multi-Model using eight generic energy-models (labeled 0-7) and by the Meta-Model (M).

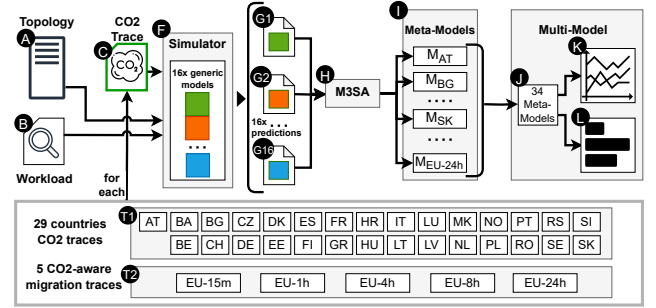


Fig. 13. Overview of experiment 3, assessing CO2 footprint of a workload run in 29 countries, and run using 5 CO2-aware migrations intervals.

The reporting features of M3SA allow for tailored evaluation and analysis. For example, when running Marconi-22, the presence of failures leads to an increase CO2 emissions by only 0.28% (Figure 12A and B). In contrast, when running Solvinity-13, which has a much longer job duration, the failures add 21.9% CO2 emissions, equivalent to 129 kg of CO2 (Figure 12C and D). Scaling linearly to the scale of Marconi HPC, consisting of 2,982 nodes, operational phenomena would add 2.56 tons of CO2. M3SA can help C-levels predict such emissions, thus reducing the risk of unexpected emissions of the operated ICT infrastructure and helping operators cope with increasingly tight climate-related regulation.

D. Evaluating M3SA on CO2-aware workload migration

ICT infrastructure accounts for 3% of global greenhouse gasses, on par with the aviation industry [50]. To address this issue, C-levels adopt CO2 reporting and instruct their organizations to adopt CO2-aware scheduling and migration policies while still respecting Sever Level Agreements [51]; in the upcoming years, regulation could require even more detailed analysis and related actions.

In this section, we analyze how M3SA copes with CO2-aware workload scheduling. As Figure 13 depicts, the experiment includes CO2 traces from 29 countries (T1) and 5 traces at different granularities, where the location with the lowest CO2 impact is chosen at given intervals (T2). As summarized by Table I, we equip M3SA with the same 16 individual energy models and ICT infrastructure in each experiment. To

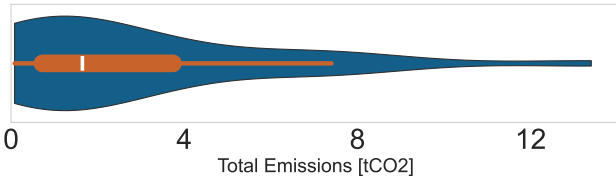


Fig. 14. Total CO2 emitted when running the Marconi-22 workload on S3, assessed for 29 European countries, as of June 2023.

enhance prediction robustness and alleviate per-model biases, we synthesize a Meta-Model for each location. Then, we run a greedy CO2-aware migration algorithm and obtain 5 CO2-aware migration traces (T2).

Overall, our results show that CO2-aware workload migration and scheduling processes can help reduce CO2 emissions significantly (MF4) and should be considered more extensively by operators during both capacity planning and operational phases.

We design and prototype a greedy CO2-aware migration algorithm (Appendix C [44], which migrates the workload to the greedy-best CO2 location at time intervals, assuming no migration costs, instant migration, and sufficient infrastructure capacity. For this experiment, we select migration intervals of 15 minutes, 1 hour, 4 hours, 8 hours, and 24 hours (T2).

We analyze the number of migrations for each month of 2023 and observe that June has the most overall migrations while January has the least. We run Marconi-22 workload, spanning over 30 days, in June 2023 and observe the CO2 footprint distribution ranging from less than 100 kg CO2 (e.g., Sweden, Switzerland) up to 13 tons CO2 (e.g., Germany), with an average of 2.92 tons CO2. The differences, some in orders of magnitude, reflect the CO2 intensity of the energy sources; for example, in Switzerland, less than 1% of electricity was obtained from brown sources [52], compared to 37% in Germany [53].

We identify migration intervals of 15 minutes and 1 hour as emitting 11% less CO2 than running the workload integrally in the lowest CO2 location, and 97.5% fewer emissions, equivalent to 2.9 tons than running the workload integrally in an average location; Figure 15 shows top 10 CO2 locations (vertical axis) and cumulated CO2 emissions (horizontal axis). Still, while 24-hour migrations are more sustainable than 26 out of 29 locations, daily migrations can result in 73% more emissions than running the workload in Switzerland, the lowest CO2 location.

The reporting features of M3SA allow for tailored CO2-aware migration and analysis and aid the migration algorithm in selecting the locations by CO2 footprint. This experiment unveils a tradeoff between migration granularity and environmental impact, unveiling new research areas in which M3SA can help identify, compare, and evaluate CO2-aware migrators and schedulers. Such tools would assist datacenter operators in making better-informed decisions about workload scheduling and migration, significantly improving operational footprint.

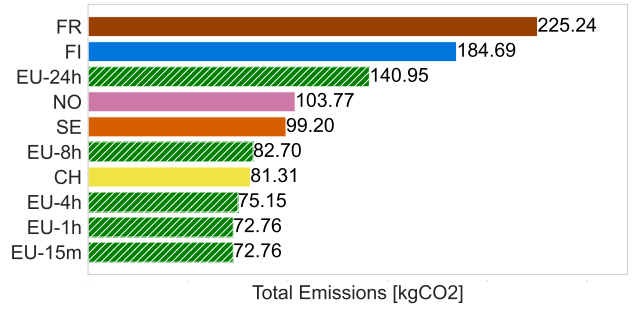


Fig. 15. Lowest 10 CO2 cumulated emissions, from lowest CO2 locations and migration models, at different granularities.

V. RELATED WORK

Simulating using multiple models is a novelty in computer systems; however, as introduced in Section I, other scientific fields already employ multi-model simulations, such as virology [19], weather and climate simulation [28], or ecology [29]. In ICT, existing simulators currently rely on single model predictions [14]–[18]. We are the first to propose Multi- and Meta- Model simulation of ICT infrastructure.

Closest to our work, we identify COVID-19 Forecast Hub [19], used to predict the COVID-19 hospitalizations in the United States, and providing 93 models, out of which users can select and leverage in a unified visualization. This is equivalent to the *Multi-Model* component of M3SA. In weather and climate simulation [20], Myhre et al. investigated the geographical distribution of harmful emissions using seven models unified under a single representation and aggregated (averaged) into a “Mean Model”, helping to contrast biases and improve prediction explainability. This is equivalent to the *Meta-Model* component of M3SA.

Datacenter simulators are useful tools for the community and are much used in predicting ICT infrastructure under workload. However, no simulator provides multi- and meta-simulation capabilities but relies on single-model predictions [14]–[18]. Our work adds this capability as an external add-on.

VI. CONCLUSION AND FUTURE WORK

Understanding the performance and climate impact of existing and, more importantly, upcoming datacenters is essential to our society and economy. Although datacenter operators already use simulators to predict infrastructure capabilities, current simulators often rely on single models, which are insufficient for reliable predictions under diverse scenarios.

In this work, we have designed, implemented, and evaluated M3SA, a framework for multi- and meta-model simulation and analysis of datacenters. We prototyped M3SA and evaluated it using real-world workload and carbon-emission traces, demonstrating M3SA’s ability to predict in complex scenarios, with enhanced explainability and robustness over approaches that only use single models (even hand-tuned). Such capabilities can aid analysts and C-level decision-makers to better reason about real-world scenarios and make informed decisions. Results show that M3SA’s can simulate years of

datacenter operation in minutes, with a 50% lower error rate than single-model-based predictions, while enabling detailed explanation and without significant overheads. We also show-case how M3SA can simulate system failures and contribute to CO₂-aware workload migration and scheduling, reducing CO₂ emissions significantly when considering datacenters' geographical location.

We have released M3SA as an open-sourced system that can be applied as a top layer to datacenter simulators and tested its operation with the commonly used open-source simulator OpenDC. We envision future research in employing AI/ML techniques in the Meta-Model aggregation function and are currently embedding M3SA into a digital twinning simulation system in a major infrastructure project with over 75 partner institutions.

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TABLE V

FORMULAS OF THE POWER MODELS USED IN THIS WORK. E=EXPERIMENT (E.G., E1=EXPERIMENT 1). P_{idle} , P_{max} = THE POWER USED IN IDLE AND FULL CAPACITY STATES, u = CPU UTILIZATION, e = EULER'S NUMBER, α = UTILIZATION FRACTION AT WHICH THE HOST BECOMES ASYMPTOTIC, r = CALIBRATION PARAMETER.

ID	Name	Formula	Source	E1	E2	E3
EQ1	Sqrt	$P(u) = P_{idle} + (P_{max} - P_{idle}) \cdot \sqrt{u}$	[14], [15], [25]	✓	✓	✓
EQ2	Linear	$P(u) = P_{idle} + (P_{max} - P_{idle}) \cdot u$	[14], [15], [25]	✗	✓	✓
EQ3	Square	$P(u) = P_{idle} + (P_{max} - P_{idle}) \cdot u^2$	[14], [15], [25]	✗	✓	✓
EQ4	Cubic	$P(u) = P_{idle} + (P_{max} - P_{idle}) \cdot u^3$	[14], [15], [25]	✗	✓	✓
EQ5	MSE	$P(u) = P_{idle} + (P_{max} - P_{idle}) \cdot (2u - u^r)$	[14], [54]	✓	✓	✓
EQ6	Asymptotic	$P(u) = P_{idle} + \frac{(P_{max} - P_{idle})}{2} \cdot (1 + u - e^{-u/\alpha})$	[14]	✓	✓	✓
EQ7	Asymptotic DVFS	$P(u) = P_{idle} + \frac{(P_{max} - P_{idle})}{2} \cdot (1 + u^3 - e^{-u^3/\alpha})$	[14]	✓	✓	✓

TABLE VI

POWER MODELS USED IN THIS WORK. E=EXPERIMENT (E.G., E1=EXPERIMENT 1). P_{idle} , P_{max} = POWER USED IN IDLE AND FULL CAPACITY STATES, α = UTILIZATION FRACTION AT WHICH THE HOST BECOMES ASYMPTOTIC, r = CALIBRATION PARAMETER.

ID	Name	P_{min}	P_{max}	r	α	E1	E2	E3
M1	Sqrt	32	180	N/A	N/A	✓	✓	✓
M2	Sqrt	0	180	N/A	N/A	✗	✗	✓
M3	Linear	32	180	N/A	N/A	✗	✓	✓
M4	Linear	0	180	N/A	N/A	✗	✗	✓
M5	Square	32	180	N/A	N/A	✗	✓	✓
M6	Square	0	180	N/A	N/A	✗	✗	✓
M7	Cubic	32	180	N/A	N/A	✗	✓	✓
M8	Cubic	0	180	N/A	N/A	✗	✗	✓
M9	Mse	32	180	10.0	N/A	✓	✗	✗
M10	Mse	32	180	0.7	N/A	✗	✓	✓
M11	Mse	0	180	0.7	N/A	✗	✗	✓
M12	Asymptotic	32	180	N/A	0.30	✓	✗	✗
M13	Asymptotic	32	180	N/A	0.85	✗	✓	✓
M14	Asymptotic	0	180	N/A	0.85	✗	✗	✓
M15	AsymptoticD	32	180	N/A	0.30	✓	✗	✓
M16	AsymptoticD	32	180	N/A	0.85	✗	✓	✓
M17	AsymptoticD	0	180	N/A	1.90	✗	✗	✓
M18	AsymptoticD	32	180	N/A	1.90	✗	✓	✓

APPENDIX A - EXPERIMENT OVERVIEW

In Appendices A - D, we present a technical report, complementary to the experiments run in Section IV. In Table V, we give an overview of the models used to predict the energy usage of the infrastructure under workload, the formulas used by the models, relevant work embedding these models, and the experiments where they have been used. In Table VI, we give an overview on the model configuration used throughout the experimentation process.

APPENDIX B - EXPERIMENT 1 - PERFORMANCE ANALYSIS

In this section, we present complementary technical information on the performance evaluation of the M3SA system. In

Section IV-B, we run M3SA using four individual models in the simulation and analysis process. We present in Table VII the time it took OpenDC to simulate using four individual models and the time it took M3SA to leverage, compute, and plot a Multi- and Meta-Model.

We ran the experiment on a regular-user machine (i.e., not a supercomputer): a MacBook Pro 16 with an M2 Pro chip, 16 GB RAM, MacOS Sequoia 15.0.1, and no user activities running in the background. We ran each experiment 10 times without changing the experimental setup. Although dependent on the machine, operating system, et cetera, we expect similar proportions between the simulation time and M3SA overhead on other machines, allowing M3SA to run years of real-world datacenter operation within minutes.

TABLE VII

M3SA PERFORMANCE EVALUATION (NFR1). σ = STANDARD DEVIATION, OVER 10 SAMPLES. RWO = REAL-WORLD OPERATIONAL TIME.

Samples	RWO	Simulation Time	Time σ	M3SA Overhead	Time σ	NFR1
2,016	7 days	5.26 s	0.56	1.35 s	0.33	26 %
4,032	14 days	8.82 s	1.09	1.74 s	0.29	19 %
10,080	35 days	16.59 s	0.71	2.05 s	0.21	12 %
20,160	70 days	30.83 s	0.82	3.68 s	0.23	12 %
50,400	175 days	73.31 s	1.07	10.91 s	0.48	15 %
100,800	350 days	147.11 s	1.76	23.71 s	1.97	16 %
201,600	700 days	290.40 s	5.09	50.72 s	1.82	17 %
403,200	1400 days	582.76 s	26.09	111.43 s	10.31	19 %

APPENDIX D - EVALUATING M3SA ON CO2-AWARE WORKLOAD MIGRATION

In this section, we extend Section IV-D and present complementary technical details on CO2-aware scheduling and workload migration using M3SA capabilities.

We analyzed the capabilities of M3SA in CO2-aware workload scheduling and migration. In the experiment, we identified that M3SA can help C-level stakeholders with CO2-aware workload scheduling, as well as increase the prediction reliability and significantly reduce CO2 predictions. We designed

TABLE VIII
MIGRATION COUNT AT VARIOUS INTERVALS, FOR EACH MONTH OF THE
YEAR 2023, USING ENTSGE-EU-23 CARBON TRACE.

Month	15 min	1 h	4 h	8 h	24 h
Jan	4	4	4	2	2
Feb	24	24	20	2	2
Mar	73	73	33	11	3
Apr	43	43	25	19	7
May	17	17	6	0	0
Jun	112	112	69	35	11
Jul	105	105	46	27	7
Aug	112	112	58	28	8
Sep	30	30	21	6	0
Oct	112	111	53	32	8
Nov	62	62	28	15	6
Dec	53	53	16	8	3

and engineered a greedy CO₂-aware migration algorithm, which underlies the pseudocode from Section VI and migrates the workload to the lowest CO₂ intensity location, assuming no migration cost.

After establishing the experimental setup, we perform the experimental process. In Section VI, we use Marconi-22, a research-specific workload containing 8,316 jobs spanned over 30 days. We leverage the number of migrations for each month and synthesize the results in Table VIII. Then, in Figures Figure 19-30, we visually represent the CO₂ emissions per country for each month of 2023.

Evaluating the migration count for 2023, we observe that June contains the highest number of overall migrations, while January contains the least. Despite June 2023 having the most overall migrations for all the evaluated migration granularities, we observe that the other two summer months, July and August, and the mid-autumn month of October, have a similar magnitude number of migrations. The number of migrations directly depends on the energy source and, thus, weather conditions (e.g., sun, wind, river flow). This per-month evaluation of the number of migrations helped us select June 2023 as the month to run the workloads.

```
def migrate_at_granularity(metamodels, granularity){
    current = lowest_co2_at_timestamp(metamodels, 0)
    model_length = len(metamodels[0].timestamps)
    migrations = 0

    for (i = 0; i < model_length; i++){
        if (i % granularity == 0) {
            best = lowest_co2_at_timestamp(metamodels, i)

            if (current.location == best.location) {
                current = best
                migrations++
            }
        }
    }
}
```

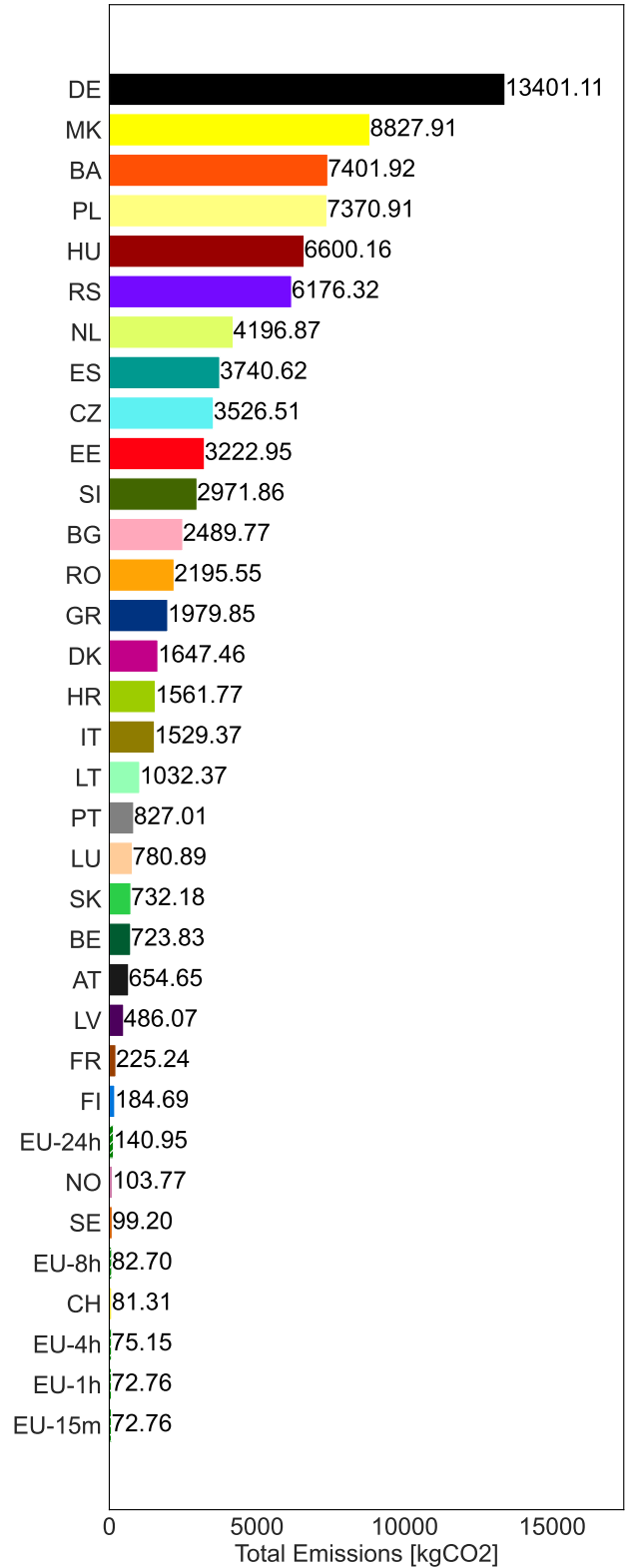


Fig. 16. Total CO₂ emissions of S3 infrastructure, run in 29 different locations, and compared to 5 migration granularities. Plotted at linear scale.

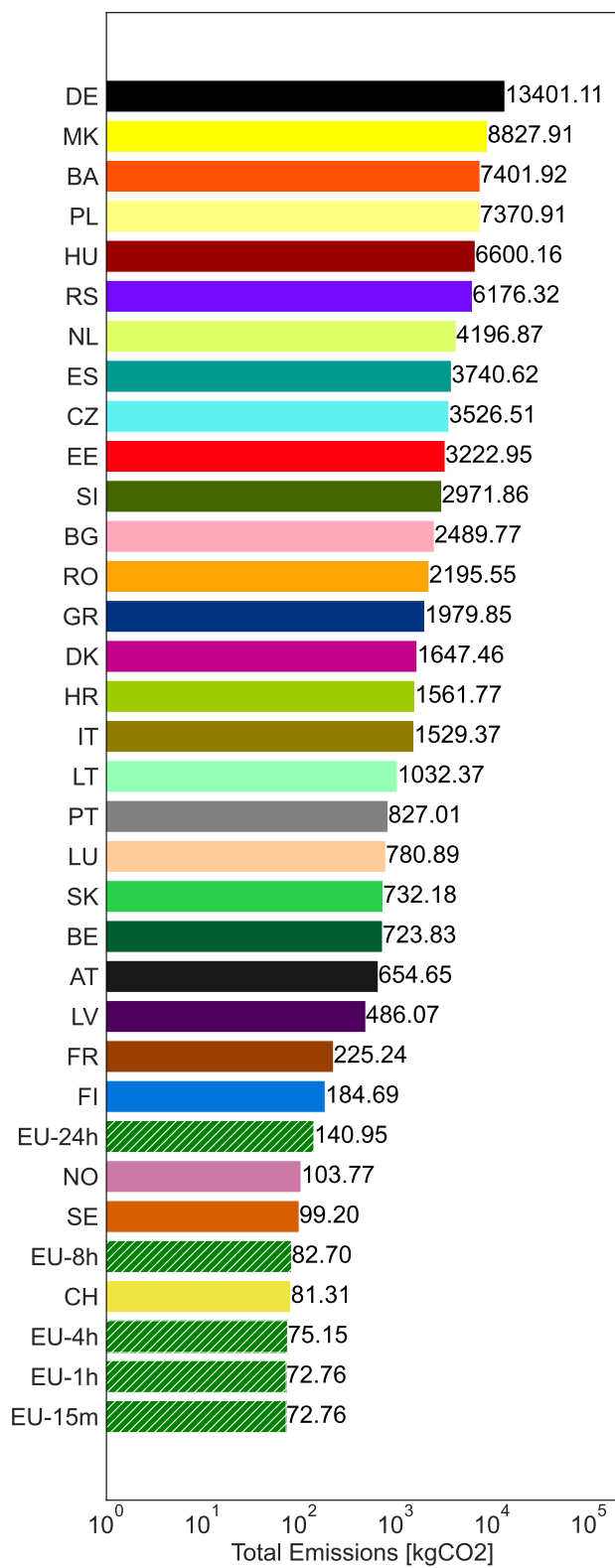


Fig. 17. The homonym of Figure 16, plotted at logarithmic scale.



Fig. 18. Legend for Figures 19-30. The colors are kept constant for the corresponding countries.

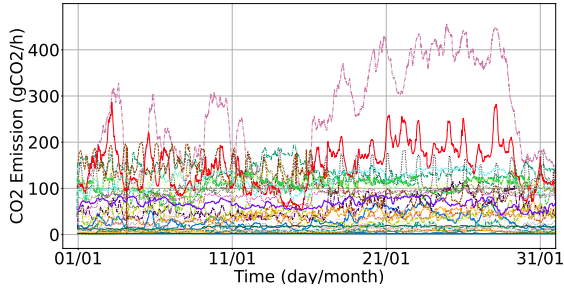


Fig. 19. CO2 emissions Jan 2023.

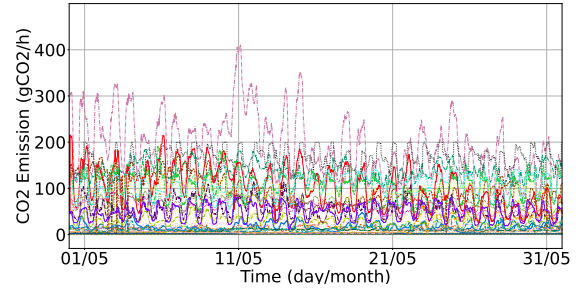


Fig. 23. CO2 emissions May 2023.

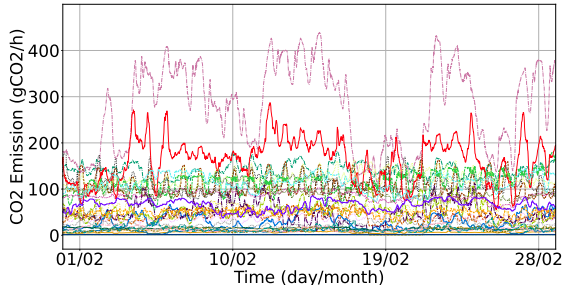


Fig. 20. CO2 emissions Feb 2023.

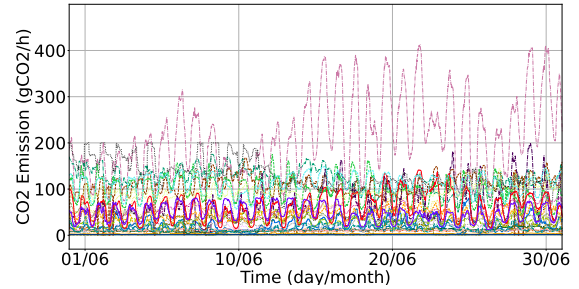


Fig. 24. CO2 emissions Jun 2023.

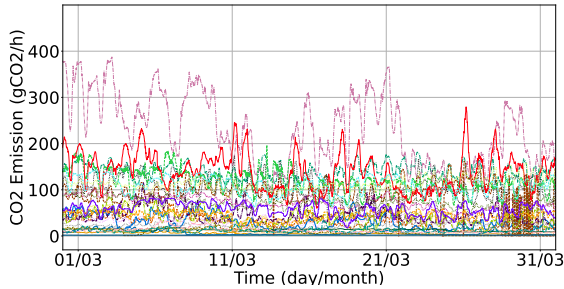


Fig. 21. CO2 emissions Mar 2023.

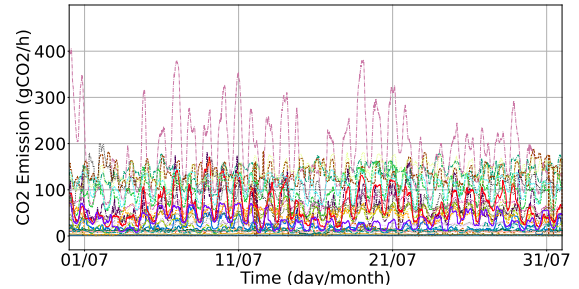


Fig. 25. CO2 emissions Jul 2023.

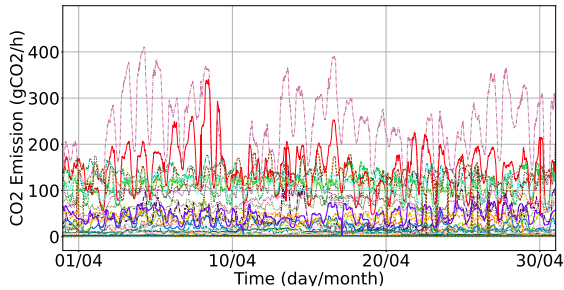


Fig. 22. CO2 emissions Apr 2023.

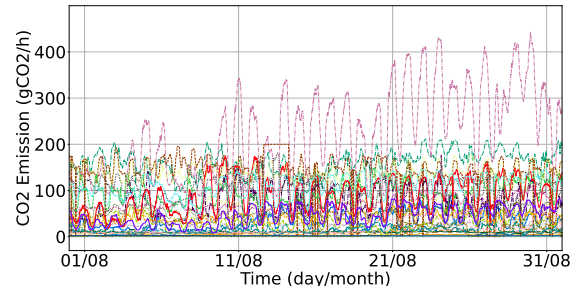


Fig. 26. CO2 emissions Aug 2023.

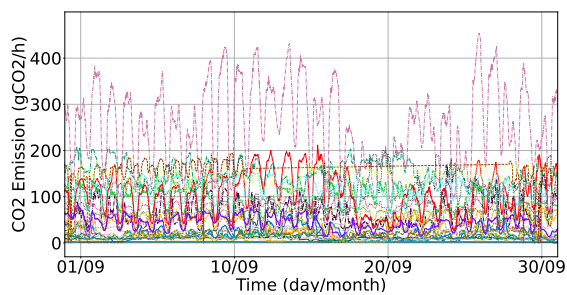


Fig. 27. CO2 emissions Sep 2023.

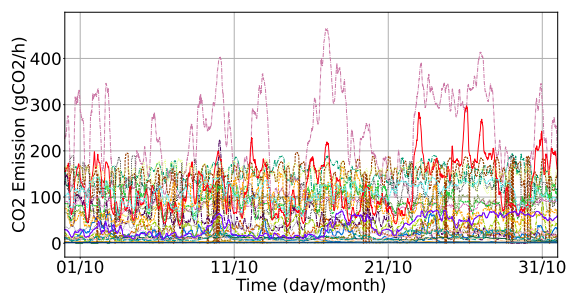


Fig. 28. CO2 emissions Oct 2023.

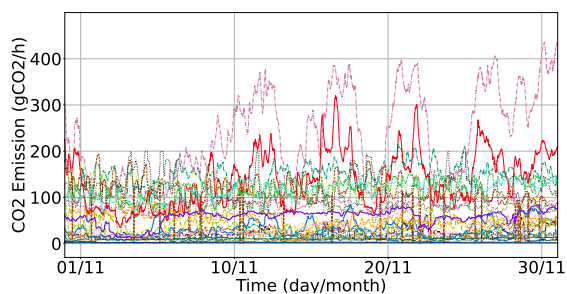


Fig. 29. CO2 emissions Nov 2023.

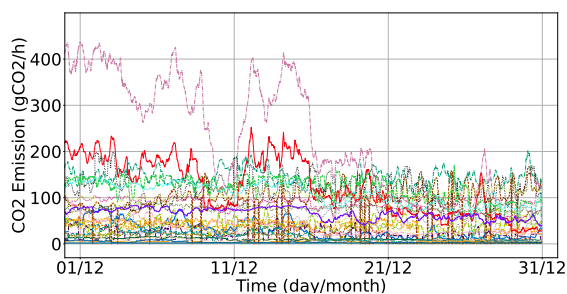


Fig. 30. CO2 emissions Dec 2023.